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Inventor(s)	Ogawa; Takeshi

Signal processing apparatus and non-transitory computer-readable storage medium

Abstract

This invention provides a signal processing apparatus which comprises a CPU and a programmable signal processor which has a plurality of execution units, a register file having a plurality of registers connected serially, and a shift controller which issues a shift signal to shift data held in the register file when each of the execution units has completed execution of a program of one cycle. Each of the execution units, after the execution of the program of one cycle, re-executes the program in accordance with the reception of the shift signal. The CPU stores, to a program memory in each of the execution units, a program including instructions to store data of a result of processing to one of registers so that the other execution unit refers the data in next cycle.

Inventors:	Ogawa; Takeshi (Tokyo, JP)
Applicant:	CANON KABUSHIKI KAISHA (Tokyo, JP)
Family ID:	89169857
Assignee:	CANON KABUSHIKI KAISHA (Tokyo, JP)
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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5481746	12/1995	Schiffleger	712/E9.034	G06F 15/8084
7546444	12/2008	Wolrich	712/228	G06F 9/30127
11977509	12/2023	Master	N/A	G06F 5/01
2006/0020768	12/2005	Toda	712/E9.063	G06F 9/3867
2017/0322813	12/2016	Langhammer	N/A	G06F 9/3869

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
4222808	12/2008	JP	N/A
2011-089913	12/2010	JP	N/A

Primary Examiner: Faherty; Corey S

Attorney, Agent or Firm: Cowan, Liebowitz & Latman, P.C.

Background/Summary

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a signal processing apparatus and a non-transitory computer-readable storage medium.

Description of the Related Art

(2) Conventionally, there have been programmable signal processing apparatuses (signal processors) that dynamically configure signal processing circuits such as Field-Programmable Gate Arrays (FPGA) or reconfigurable circuits, and that execute programs that sequentially execute instruction sequences as with a DSP.

(3) For example, a method of cascade-connecting DSPs executing different programs to form a multi-stage configuration is disclosed in Japanese Patent No. 4222808 (hereinafter referred to as “Document 1”). Since the programs are different, the time required for processing by each DSP will also differ among the DSPs. Therefore, in order to improve the throughput of the DSPs, a contrivance is required to prevent the operation rate of the DSPs from decreasing due to an effect

of BUSY states of the other DSPs that operate cooperatively therewith.

(4) In Japanese Patent Laid-Open No. 2011-89913A (hereinafter referred to as “Document 2”), DSPs that execute different programs are configured to calculate data that they have each read from a memory by way of a Direct Memory Access Controller (DMAC), and then write the calculation result into the memory. Also, Document 2 discloses a method of IPC (inter-process communication) for synchronization between DSPs for realizing a data flow equivalent to a configuration in which DSPs are connected in series, by the above-described configuration.

(5) However, in the conventional technique disclosed in the above-mentioned Document 2, while the operation rate of the DSPs is improved, an interface circuit between the DSPs, a control CPU, a control program thereof, or the like are required, which increases the circuit size of the entire signal processing apparatus.

SUMMARY OF THE INVENTION

(6) The present invention has been made in view of the above problems, and provides a technique for simplifying a structure related to data transfer between program execution units that operate in parallel in a programmable signal processing circuit.

(7) According to an aspect of the invention, there is provided a signal processing apparatus comprising: a CPU; and a programmable signal processing circuit having the following: a plurality of execution circuits each of which has a program memory and executes a program stored in the program memory, wherein the plurality of execution circuits can operate in parallel; a register file used for the plurality of execution circuits, wherein the register file has a plurality of registers serially connected and transfers, in accordance with a shift signal, data held by each of the plurality of registers to a register located downstream; a shift controller that issues the shift signal to the plurality of execution circuits at a timing at which the plurality of execution circuits do not execute the program, wherein the CPU stores, to each program memory of the plurality of execution circuits, a program including instructions to make a register of the register file located upstream of a register that another execution circuit is to reference in order to input data be a storage destination for data of a result by the execution of the program so that the data of the result by the execution of the program by the execution circuit is transferred to the another execution circuit via the register file.

(8) According to the present invention, it is possible to simplify a structure related to data transfer between program execution units that operate in parallel in a programmable signal processing circuit.

(9) Further features of the present invention will become apparent from the following description of exemplary embodiments (with reference to the attached drawings).

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a circuit configuration diagram of a programmable signal processor according to an embodiment.

(2) FIG. 2 is a circuit configuration diagram of a shift register group in a register file unit in an embodiment.

(3) FIG. 3 is a circuit configuration diagram of a first processing unit in an embodiment.

(4) FIGS. 4A and 4B are views illustrating bit assignments of instructions executed by a first processing unit according to an embodiment.

(5) FIG. 5 is an equivalent circuit diagram of a time when each execution unit of a programmable signal processor executes a program according to an embodiment.

(6) FIG. 6 is view illustrating a list of programs that each execution unit of the programmable signal processor will execute according to an embodiment.

- (7) FIG. 7 is a timing chart diagram for describing processing of the programmable signal processor according to the embodiment.
- (8) FIGS. 8A to 8C are views for comparing each method related to data transfer.
- (9) FIG. 9 is a view illustrating a memory map of an IO bus according to an embodiment.
- (10) FIG. 10 is a block diagram of an electronic device according to an embodiment.
- (11) FIG. 11 is a flowchart illustrating a processing sequence of a CPU of an embodiment.
- (12) FIG. 12 is a circuit configuration diagram of a programmable signal processor according to a second embodiment.
- (13) FIG. 13 is a view illustrating a program list to be executed by the first processing unit of the second embodiment.
- (14) FIG. 14 is a circuit configuration diagram of a programmable signal processor according to a third embodiment.
- (15) FIG. 15 is a block diagram of an electronic device according to a fourth embodiment.
- (16) FIG. 16 is a view illustrating a program list to be executed by the first processing unit of the fourth embodiment.
- (17) FIG. 17 is a view illustrating bit assignments of floating point data according to the fourth embodiment.

(18) FIG. 18 is an equivalent circuit diagram of a signal processor of the fourth embodiment.

DESCRIPTION OF THE EMBODIMENTS

- (19) Hereinafter, embodiments will be described in detail with reference to the attached drawings. Note, the following embodiments are not intended to limit the scope of the claimed invention. Multiple features are described in the embodiments, but limitation is not made to an invention that requires all such features, and multiple such features may be combined as appropriate.
- (20) Furthermore, in the attached drawings, the same reference numerals are given to the same or similar configurations, and redundant description thereof is omitted.

First Embodiment

- (21) FIG. 10 illustrates a block diagram of an information processing apparatus according to a first embodiment. For ease of understanding, in the first embodiment, an example in which the information processing apparatus is applied to an image capturing apparatus typified by a digital camera, for example, will be described.
- (22) The information processing apparatus of the first embodiment includes a programmable signal processor **1001**, a Central Processing Unit (CPU) **1005**, memories **1003a** to **1003d**, a memory bus **1002**, and an IO bus **1006**. The CPU **1005** manages control of the entire information processing apparatus, and performs control of loading and activating a program to the programmable signal processor **1001**. The memories **1003a** to **1003d** are used to store programs executed by the CPU **1005** and the signal processor **1001**, as well as image data to be processed and image data after processing. The memories **1003a** to **1003d** are configured by a RAM, a ROM, or the like.
- (23) The memory bus **1002** is a multi-layer type memory bus that connects a plurality of memories (memory **1003a** to **1003d**) and a plurality of bus masters (including the illustrated signal processor **1001**, a CPU **1005**) and transfers data. The IO bus **1006** is an IO bus that connects a plurality of devices and a controller (such as a signal processor **1001**, the CPU **1005**) and transfers data.
- (24) Note, FIG. 10 shows only the main portion according to the present embodiment. For example, in a case where the configuration of FIG. 10 is applied to an image capturing apparatus typified by a digital camera, an image capturing unit for obtaining image data to be processed, a recording unit for recording processed image data in a nonvolatile memory or the like, and a user interface (operation unit, display unit) between the user and the present apparatus are connected to the IO bus **1006** as the above-described device.
- (25) The CPU **1005** controls the entire apparatus by executing programs stored in the memories **1003a** to **1003d** via the memory bus **1002**. The CPU **1005** also controls the signal processor **1001** through the IO bus **1006**. Specifically, the CPU **1005** reads a plurality of programs executed by the

signal processor **1001** from the memories **1003a** to **1003d**. The CPU **1005** writes the read programs to a plurality of memories (reference numerals **102**, **150**, **111**, and **113** to be described later) built into the signal processor **1001** via IO bus **1006**, instructs the signal processor **1001** to activate, and executes processing.

(26) Upon receiving an instruction to activate from the CPU **1001** via the IO bus **1006**, the signal processor **1001** reads the image data to be processed stored in the memories **1003a** to **1003d**, executes filter processing, and writes the filter-processed image data into the memories **1003a** to **1003d**.

(27) Next, the configuration of the signal processor **1001** according to the embodiment will be described in more detail. FIG. **1** is a circuit configuration diagram of the signal processor **1001** according to the embodiment.

(28) As illustrated, the signal processor **1001** includes a reading unit **101**, a setter **104**, a first processing unit **110**, a second processing unit **112**, a writing unit **114**, a register file unit **106**, and a shift control unit **117**. Each of the reading unit **101**, the setter **104**, the first processing unit **110**, and the second processing unit **112** incorporates a memory for holding a program to be executed by itself. Reference numerals **102**, **105**, **111**, and **113** of the illustration are these memories. As described above, the CPU **1005** stores programs (four programs) for the signal processor **1001** read from the memories **1003a** to **1003d** into the memories **103**, **105**, **111**, and **113**, respectively, via the IO bus **1006**. The CPU **1005** also performs processing for setting a required value for the signal processor **1001** in a control register held by the signal processor **1001**.

(29) The register file unit **106** is configured by a general-purpose shift register group **107**, a general-purpose register group **108**, and an output register **109**. Each of the setter **104**, the first processing unit **110**, the second processing unit **112**, and the writing unit **114** can directly use each register within the register file unit **106**.

(30) The general-purpose shift register group **107** is configured by a plurality of shift registers connected in series. Each shift register can hold 32 bits of data. Then, in response to a shift cycle signal from the shift control unit **117**, each shift register transfers the data held by itself to a shift register one downstream (details will be described later). Note, although there is no particular limitation on the number of shift registers constituting the general-purpose shift register group **107** in the embodiment, it is assumed that the configuration is made with 16 shift registers in the embodiment. When expressed by operands in a program executed by the signal processor **1001**, these serially connected shift registers are denoted by **R00**, **R01**, **R02**, . . . , **R0e**, and **R0f** from the most upstream to the downstream (the two characters on the right are expressed in hexadecimal).

(31) The general-purpose register group **108** is configured by a plurality of registers. Although there is no particular limitation on the number of registers, the number of registers is 15 in the embodiment. When expressed by operands in a program, the registers constituting the general-purpose register **108** are denoted by **R10**, **R11**, **R12**, . . . , and **R1e**. Unlike the shift registers within the general-purpose shift register **107**, these registers hold data (32 bits) independently of a shift cycle signal.

(32) The output register **109** is a register for holding filter-processed data (32 bits). It is assumed that the output register **109** is expressed as “**R1f**” in operand notation.

(33) As described above, the register file unit **106** in the embodiment has a total of 32 registers.

(34) The reading unit **101** reads image data to be processed within the memories **1003a** to **1003d** in units of pixels by executing a program stored in the program memory **102** internally, and then stores the image data into the memory **103**, which is a First In First Out (FIFO) memory. As described above, since processing of the reading unit **101** is a relatively simple process of reading the image data from a specified address and storing the image data into the FIFO memory **103**, a program expressing a process of updating the read address is stored in the program memory **102**. Note, in a case where there is free space in the FIFO memory **103**, the reading unit **101** executes read processing. Also, in a case where the FIFO memory **103** is in a memory full state in which

there is no free space, the reading unit **101** temporarily stops reading and waits for free space to be generated.

(35) The setter **104**, by executing a program stored in the program memory **105**, performs processing for inputting image data from the FIFO memory **103** and setting (or assigning) the image data to a corresponding register in the register file unit **106**. That is, the program memory **105** stores a program indicating to which register the input image data is to be set. Note that when the setter **104** inputs data from the FIFO memory **103**, a free space for one data unit is generated in FIFO memory **103**.

(36) The first processing unit **110** executes a program stored in the program memory **111**. The first processing unit **110** performs processing using registers in the register file unit **106** when performing various calculation processing according to the program.

(37) The second processing unit **112** executes a program stored in the program memory **112**. The second processing unit **112** performs processing using registers in the register file unit **106** when performing various calculation processing according to the program. Also, the second processing unit **112** stores the image data of the final filter processing result into the output register **109** (shift register “R1f” in the program).

(38) The writing unit **114** stores data held in the output register **109** within the register file unit **106** into a write FIFO memory **116**. Then, the writing unit **114** sets the data stored in the FIFO memory **116** in a write buffer **115**. The writing unit **114** reads data from the buffer **115** and writes the data to preset addresses of the memories **1003a** to **1003d**.

(39) The configuration of the signal processor **1001** according to the embodiment has been described above. The FIFO memory **103** is interposed between the reading unit **101** and the setter **104** to absorb differences in processing timings therebetween. Note, it is assumed that the FIFO memory **103** has a capacity capable of storing three pixels worth of image data.

(40) On the other hand, there is no interface circuit for inputting and outputting data between the setter **104**, the first processing unit **110**, and the second processing unit **112**. Instead, the general-purpose shift register group **107** in the register file unit **106** plays a role of transferring data between the setter **104**, the first processing unit **110**, and the second processing unit **112** (details will be described later).

(41) FIG. 2 shows a circuit diagram of the general-purpose shift register group **107** in the register file unit **106**.

(42) Reference numeral **201** denotes a shift register located most upstream in the serial connection, which is “R00” when indicated by an operand. Reference numeral **214** denotes a shift register located at the second most upstream position of the serial connection, which is “R01” when indicated by an operand. Thereafter, the operands of the registers serially connected will continue with R02, R03, . . . , and R0f. Each register has the same structure.

(43) Here, the shift register **201** will be described. Reference numeral **202** denotes one of the flip-flops holding 32 bits of data. Reference numeral **204** denotes an input signal for transferring data at a time of a shift operation, but 0 is inputted as the input signal since the shift register **201** serves as a register at the head of the shift operation. Note, the number of bits held by the shift register **201** is not particularly limited, and may be 8 bits.

(44) Reference numeral **205** denotes an output of the flip-flop **202**, and during a shift operation, the output **205** is loaded into the shift register **214** located downstream.

(45) Reference numeral **210** denotes an input signal from the setter **104**, reference numeral **211** denotes an input signal from the second processing unit **112**, reference numeral **212** denotes an input signal from the first processing unit **110**, and reference numeral **213** denotes an input signal from the CPU **1005** outside of the signal processor **1001**.

(46) Reference numeral **203** denotes a signal for loading data into the flip-flop **202** and the other flip-flops in the shift register at the time of a shift operation. Reference numeral **206** denotes a signal that is outputted when the setter **104** writes the input signal **210** to the flip-flop **202**.

Reference numeral **207** denotes a signal that is outputted when the second processing unit **112** writes the input signal **211** to the flip-flop **202**. Reference numeral **208** denotes a signal that is outputted when the first processing unit **110** writes the input signal **212** to the flip-flop **202**. Reference numeral **209** denotes a signal that is outputted when the external CPU **1005** writes the input signal **213** to the flip-flop **202**.

(47) In a case where each of the reference signals **203**, **206**, **207**, **208**, and **209** is active, a switch controlled by each reference signal is connected to the lower side in the figure and outputs each input signal **210**, **211**, **212**, and **213** to the flip-flop **202**. Also, as can be seen from the figure, in a case where a load signal of each of the reference signals **203**, **206**, **207**, **208**, and **209** is not active, the switch controlled by each reference signal is connected to the upper side in the figure. In a case where each of the reference signals **203**, **206**, **207**, **208** and **209** is not active, the value of the flip-flop **202** returns to the flip-flop **202** again, so that the value of the flip-flop **202** is held even if a clock is supplied.

(48) In addition, there is no circuit for arbitrating load processing of data by each of the reference signals **203**, **206**, **207**, **208** and **209**. Therefore, in a case where a plurality of reference signals become active at the same time, the data selected by the switch closest to the flip-flop **202** is loaded into the flip-flop **202**. However, in the embodiment, a plurality of processing units are configured so as to not write to one shift register simultaneously.

(49) In this way, the setter **104**, the first processing unit **110**, and the second processing unit **112** in addition to the CPU **1005** can set to and can also reference (read) the shift register **201**.

(50) For example, the setter **104** can set (write) data from the FIFO memory **103** to the register **201** by outputting the data to be written to a signal line **210** and then activating the signal **206** for writing. Further, a first signal processing unit outputs the data to be written to the signal **212** and then activates the signal **208**, so that the first signal processing unit can write the data to the register **201**.

(51) FIG. 3 is an internal block diagram of the first processing unit **110**. Reference numeral **301** denotes a program counter indicating an instruction position to be executed by a program in the program memory **111**. Reference numeral **303** denotes an instruction selection multiplexer for retrieving an instruction at a position indicated by the program counter **301** in the program memory **111**. Reference numeral **304** denotes a register bus which bundles the output signals of all the flip-flops of the register file unit **106**. Reference numeral **305** denotes a multiplexer that selects a signal of a first source, which is an input signal to be calculated, from the register bus **304**. Reference numeral **306** denotes a multiplexer that selects a signal of a second source, which is an input signal to be calculated, from the register bus **304**. Reference numeral **307** denotes an Arithmetic and Logic Unit (ALU) that performs a calculation according to an execution instruction **308** on the two values of the first source and the second source. Reference numeral **310** denotes a calculation result from the ALU **307**. Reference numeral **309** is a binary decoder that activates a load signal **311** to any register of the register file unit **106** in accordance with the execution instruction **308**. Reference numeral **312** denotes an END detection unit which detects an instruction "END" indicating the end of storage in the program memory **111**. The END detection unit **312** outputs a signal **313** in a case where an END instruction is detected. The program counter **301** stops updating the program counter (address) in response to the signal **313**, and sets the value to the head of the program. The END detection unit **312** also outputs the signal **313** to the shift control unit **117**.

(52) The structure of the first processing unit **110** has been described above. The second processing unit **112** has the same structure as that of the first processing unit **110**, and description thereof will be omitted. Also, the setter **104** has a program counter and an END detection unit, and when the END detection unit detects an END instruction, the address updating is stopped and the value is set to the head of the program.

(53) The shift control unit **117** may receive a detection signal of an END instruction from each of the setter **104**, the first processing unit **110**, and the second processing unit **112**. In a case where the

shift control unit **117** receives the detection signal of the END instruction from all of the setter **104**, the first processing unit **110**, and the second processing unit **112**, it activates the signal **203** (with reference to FIG. 2) and issues a shift cycle signal. As a result, each shift register constituting the general-purpose shift register group **107** transfers data to a shift register one downstream. In response to the shift cycle signal, the setter **104**, the first processing unit **110**, and the second processing unit **112** start processing for the next cycle.

(54) FIGS. 4A and 4B show a program instruction stored in the memory **111** of the first processing unit **110** (or the memory **113** of the second processing unit).

(55) The FIG. 4A illustrates the roles of bits in a 19-bit instruction. The instruction consists of four fields. The 4 bits of bits **18** to **15** are an opcode field **401** indicating the type of calculation (addition, subtraction, logical operation, etc.), that is, an operator. The 5 bits of bits **14** to **10** are a destination field **402** indicating the storage destination for storing the calculation result. The 5 bits from bit **9** to bit **5** are a second source field **403** indicating the second source. Also, the 5 bits from bit **4** to bit **0** are a first source field **404** indicating the first source. In this way, a role is determined for each bit in the instruction.

(56) FIG. 4B shows an exemplary mnemonic notation on the assembler of these instructions.

(57) Reference numeral **405** denotes an opcode indicating “addition”, reference numeral **406** denotes a setting destination register, and reference numerals **407** and **408** denote reference registers. The opcode **405** corresponds to the opcode field **401** and the setting destination register **406** corresponds to the destination field **402** of FIG. 4A. The reference register **407** corresponds to the second source field **403**, and the reference register **408** corresponds to the first source field **404**.

(58) An operation performed in a case where the first processing unit **110** executes the instruction shown in FIG. 4B is as follows. The first processing unit **110** causes the multiplexer **305** to select the shift register **R00** in the register file unit **106** and then causes the multiplexer **306** to select the shift register **R01**. Then, the ALU **307** adds values selected by the multiplexer **305** and **306** (the values read from each of the shift registers **R00** and **R01**) in accordance with the execution instruction **308** (the opcode “ADD” in FIG. 4B), and then outputs the calculation result **310**. At this time, the decoder **309** selects the shift register “**R02**” in the destination field **402** and then activates the signal **208**. As a result, the result of adding the two values held in the shift register **R00** and **R01** is written to the shift register **R02**.

(59) FIG. 9 shows the memory space of the IO bus **1006** of the signal processor **1001** in the embodiment.

(60) The data bus of the IO bus **1006** is 32-bit and is readable and writable from the CPU **1005**. The shift registers **R00**, **R01**, and . . . constituting the general-purpose shift register group **107** are allocated from a base address **0x0000** indicated by reference numeral **901** to an address **0x003C** indicated by reference numeral **902** (an address starting with “0x” is represented in hexadecimal notation). Also, the registers **R10**, **R11**, and . . . constituting the general-purpose register group **108** are allocated from an address **0x0040** indicated by reference numeral **903**. Further, the output register **109** is allocated to an address **0x007C** indicated by reference numeral **904**.

(61) Addresses **0x0100** to **0x01ff** indicated by reference numeral **905** are allocated to the memory **111** of the first processing unit **110**. Addresses **0x0200** to **0x02ff** indicated by reference numeral **906** are allocated to the memory **113** of the second processing unit **112**. Addresses **0x0300** to **0x03ff** indicated by reference numeral **907** are allocated to the memory **102** of the reading unit **101**. Addresses **0x0400** to **0x04ff** indicated by reference numeral **908** are allocated to the memory **105** of the setter **104**. Also, reference numeral **909** is allocated to a control register of the signal processor **1001**. The control register includes, for example, a register describing a timing at which a series of processing ends. For example, the signal processor **1001** can be set so as to stop in a case where a processing result corresponding to a predetermined number of pixels in the horizontal direction has been obtained.

(62) As understood from FIG. 9, the memories storing each register in the register file unit **106** and

programs that each program execution unit including the reading unit **101**, the setter **104**, the first processing unit **110**, the second processing unit **112**, and the writing unit **114** have are mapped to different regions in the memory space of the IO bus **1006**. Therefore, the CPU **1005** can freely access the memories of each register in the register file unit **106** and each program execution unit. For example, in a case where the CPU **1005** writes a program of the reading unit **101** in the signal processor **1001**, the program may be written from the position of the address 0x0300. In addition, the CPU **1005** can also write an arbitrary value to an arbitrary register of the register file unit **106** by specifying the corresponding address.

(63) The signal processor **1001** has the memories **102**, **105**, **111**, and **113** that are SRAM which can be accessed at high speed. SRAM is more quickly accessible than DRAM, but is more costly per capacity than DRAM. Since each of the memories **102**, **105**, **111**, and **113** stores a relatively small program, the capacity of each memory can be reduced. Therefore, in the embodiment, SRAM is adopted as the memories **102**, **105**, **111**, and **113**, and a program can be read and written at high speed while reducing an increase in cost. Note that the memories **102**, **105**, **111**, and **113** may be configured by flip-flops.

(64) FIG. **11** shows a processing sequence related to an activation of the signal processor **1001** among programs executed by the CPU **1005**.

(65) In step **S1101**, the CPU **1005** starts the processing.

(66) In step **S1102**, the CPU **1005** reads, from the memories **1003a** to **1003d** via the memory bus **1002**, programs (four in the embodiment) to be executed by the reading unit **101**, the setter **104**, the first processing unit **110**, and the second processing unit **112**, respectively, in the signal processor **1001**. Then, The CPU **1005** writes, according to the memory map of FIG. **9** described previously, the program to be executed by the reading unit **101** with 0x0300 as the head address in the IO address space. That is, the CPU **1005** stores the program executed by the reading unit **101** in the memory **102**. Similarly, the CPU **1005** writes the program for the setter **104** into the memory **105**. Also, the CPU **1005** writes the program for the first processing unit **110** into the memory **111**. Further, the CPU **1005** writes the program for the second processing unit **112** into the memory **113**. The CPU **1005** also writes various types of data into control registers and general-purpose registers of the signal processor **1001** as needed.

(67) In step **S1103**, the CPU **1005** sets the head address of the image data in the address space of the memory bus **1002** to the reading unit **101**, and also sets the head address at a time of writing the filter-processed pixel data to the writing unit **114**.

(68) In step **S1104**, the CPU **1005** instructs the signal processor **1001** to activate and starts the processing.

(69) Then, the CPU **1005** waits for an interrupt signal indicating the end of the processing from the signal processor **1001** in step **S1105**. In a case where the CPU **1005** receives the interrupt signal indicating the end of the processing, the processing ends in step **S1106**.

(70) Once the CPU **1005** starts the process of the signal processor **1001** in this way, the end thereof need only be waited for since it is not necessary adjust the synchronization in the internal parallel operation. Note that the CPU **1005** may perform other processing while waiting.

(71) FIG. **6** shows an example of a program executed by the signal processor **1001**. Also, FIG. **5** shows an equivalent circuit diagram when the signal processor **1001** executes the program shown in FIG. **6**.

(72) Further, it is assumed that for the image **501** to be processed in the embodiment, one screen is configured by 640 pixels in the horizontal direction and 480 pixels in the vertical direction, and one pixel is represented by 32 bits. Normally, image data is often represented by 8 bits, but in this case, the image data may be processed by using 8 bits out of 32 bits. It is assumed that the data of each pixel constituting the image data to be processed are stored in order in a raster scan order from a predetermined address in the address space in the memory bus **1002**.

(73) A LIST **601** of FIG. **6** shows a program stored in the memory **102** of the reading unit **101**. A

LIST 602 shows a program stored in the memory 105 of the setter 104. A LIST 603 shows a program stored in the memory 111 of the first processing unit 110. Also, a LIST 604 shows a program stored in the memory 113 of the second processing unit 112.

(74) The reading unit 101 reads the image data from the head address position set by the CPU 1005, and then stores the read image data into the FIFO memory 103. Then, the reading unit 101 performs processing of adding the value indicated by the program LIST 601 to the previously used address, updates the read address, reads the image data from the address position after the update, and then stores the read address in the FIFO memory 103. Hereinafter, the processing of updating addresses and storing image data into the FIFO memory 103 is performed until the FIFO memory 103 is in a memory full state. Free space in the FIFO memory 103 is generated when the setter 104 located downstream performs a process of obtaining data from the FIFO memory 103.

(75) As described above, in the embodiment, the image data are stored in a raster scanning order from the predetermined address positions in the memories 1003a to 1003d. The size of the image data is 640 pixels in the horizontal direction. Therefore, the reading unit 101 will update the readout address by adding 640, 640, and -1279 offsets in order to the initial address, as shown in the LIST 601. Therefore, when the processing is started, the reading unit 101 first reads out three vertically aligned pixels including the pixels in the upper left corner of the image data and then stores the three vertically aligned pixels into the FIFO memory 103. Next, the reading unit 101 reads out the data of three pixels in the vertical direction at a position shifted by one pixel in the horizontal rightward direction, and then stores the data into the FIFO memory 103. Thereafter, the reading unit 101 repeats this processing.

(76) The setter 104 inputs image data from the FIFO memory 103 and then stores the image data into a shift register specified by a program shown in the LIST 602. It should be understood that "LOAD R00" in the LIST 602 is an instruction to "load (store) the image data read from the FIFO memory 103 into the (shift) register R00". In other words, the setter 104 performs processing of storing the uppermost pixel data of the three vertically aligned pixels into the shift register R00, the middle pixel data into the shift register R02, and the lowermost pixel data into the shift register R04. Then, since step 03 in the LIST 602 is "END", the setter 104 stops the program counter, returns the program counter to the address of the head of the program, and then waits for a shift cycle signal to be issued. By continuing the above processing, the setter 104 performs input until the right end of the image while maintaining a relationship of three pixels in the vertical direction. Note that when the image data is read from the FIFO memory 103 by the setter 104, free space is generated in the FIFO memory 103. Therefore, when the FIFO memory 103 is in a memory full state and the reading unit 101 has stopped reading, the reading unit 101 resumes the read processing of subsequent image data when the setter 104 has read the image data from the FIFO memory 103.

(77) Here, when the shift control unit 117 issues a shift cycle signal indicating a shift timing to the general-purpose shift register group 107, the general-purpose register file unit 106 transfers the data held by each of the shift registers constituting the general-purpose shift register group 107 to a shift register located one downstream.

(78) The first processing unit 110 performs processing according to the program shown in the LIST 603. It should be understood that the program "ADD R06, R01, R03" is an instruction that "adds the values of the shift registers R01 and R03 and stores the result of the addition into the shift register R06." Immediately before the first processing unit 110 starts the processing according to the program shown in the LIST 603, when the shift control unit 117 issues a shift cycle signal, the pixel data held in the shift register R00 is transferred to R01, the pixel data held in the shift register R02 is transferred to R03, and the pixel data held in the shift register R04 is transferred to R05. That is, the data of the three pixels in the vertical direction stored in the shift registers R00, R02, and R04 by the reading unit 101 are transferred to the shift registers R01, R03, and R05. In addition, the first processing unit 110 re-executes the program from the head the LIST 603 in

accordance with a reception of the shift cycle signal. When steps **00** to **02** in the LIST **603** are executed, a value of $P1+2\times P2+P3$ is stored into the shift register **R06** when the values of three pixels from the top to the bottom in the vertical direction are taken as **P1**, **P2**, and **P3**.

(79) Since step **03** of the LIST **603** is “END”, the first processing unit **110** returns the processing to the head and waits for the shift cycle signal to be issued.

(80) The second processing unit **112** performs processing according to the program shown in the LIST **604**. It should be understood that “SHIFT **R1f**, **R10**, **R1d**” in step **03** of the LIST **604** is an instruction to “shift the value of the register **R10** by the number of bits indicated by the value stored in the register **R1d** and store the shift result in the register **R1f**”. The bit shift direction depends on the value of the third operand **R1d**, and is a left shift (shift toward higher bits) in the case of positive and a right shift (shift toward lower bits) in the case of negative.

(81) In the embodiment, the CPU **1005** stores the program of the LIST **604** in the memory **113** of the second processing unit **112** and sets a value “-4” to the shift register **R1e**. In other words, since the instruction for storing “-4” in the register **R1e** is not required be described in the LIST **604**, the program-size can be reduced and the processing-throughput can be improved.

(82) When the shift control unit **117** issues the shift cycle signal immediately before the second processing unit **112** starts the processing according to the program shown in the LIST **604**, the value stored in the shift register **R06** by the first processing unit **110** in the immediately preceding cycle is transferred to the shift register **R07**. Also, the shift register **R08** stores the value stored in the shift register **R06** by the first processing unit **110** two cycles earlier. Further, the shift register **R09** stores the value stored in the shift register **R06** by the first processing unit **110** three cycles earlier.

(83) Therefore, when the second processing unit **112** executes steps **00** to **02** of the LIST **604**, a value $P1+2\times P2+P3$ is stored in the shift register **R10** when the three pixels in the horizontal direction are **P1**, **P2**, and **P3**. For the first processing unit **110** a calculation for three pixels arranged in the vertical direction is performed. Therefore, when the second processing unit **112** executes the instruction of step **03** of the LIST **604**, a value obtained by right-shifting the value of the shift register **R10** by 4 bits (equivalent to dividing by 16) is stored in the output register **R1f**. In other words, the filter-processed pixel data at the center position of the 3×3 pixel block is stored into the output register **R1f**. Since step **04** of the LIST **604** is “END”, the second processing unit **112** returns the program counter to the head and waits for the next shift cycle signal to be issued.

(84) Here, a synchronization between the reading unit **101**, the setter **104**, the first processing unit **110**, and the second processing unit **112** in the signal processor **1001** will be described with reference to the timing chart of FIG. 7.

(85) Reference numeral **701** indicates a position of a pixel corresponding to the image data read by the reading unit **101** and stored in the FIFO memory **103**. 0, 640, 1280 . . . indicate offset addresses of image data rather than the values of the image data.

(86) Reference numeral **702** denotes image data written by the setter **104**. **R00**, **R02**, and **R04** indicate a destination register of the image data, not the value of the image data. One cycle later than the reading unit **101**, the setter **104** writes the data of the offset address 0 into the shift register **R00**, the data of the offset address 640 into the shift register **R02**, and the data of the offset address 1280 into the shift register **R04**.

(87) Reference numeral **704** denotes each step executed by the first processing unit **110**, and reference numeral **705** denotes each step executed by the second processing unit **112**.

(88) Reference numeral **703** denotes a shift cycle signal. The shift control unit **117** issues the shift cycle signal **703** when all the program execution units of the setter **104**, the first processing unit **110**, and the second processing unit **112** return each of their program counters to the head in their own memories (when an END instruction is detected). As a result, the shift cycle signal is issued at a timing at which the setter **104**, the first processing unit **110**, and the second processing unit **112** do not execute the program and each of the processing units of the setter **104**, the first processing

unit **110**, and the second processing unit **112** executes the program in its own memory again in accordance with the shift cycle signal.

(89) By this mechanism, the setter **104**, the first processing unit **110**, and the second processing unit **112** can operate in parallel. In addition, data transfer is possible even if there is no interface circuit for performing data transfer between the setter **104**, the first processing unit **110**, and the second processing unit **112**.

(90) A detailed description will be given with reference to FIGS. **8A** to **8C**. FIG. **8A** is a view for describing a problem in a case where the setting program and the reference program which references set data operate in parallel. For example, in a case where the first processing unit **110** processes the data written in the shift register **107** by the setter **104**, the setter **104** is the setting program side, and the first processing unit **110** is the reference program side. In addition, in a case where the second processing unit **112** processes the calculation result of the first processing unit **110**, the first processing unit **110** becomes the setting program side, and the second processing unit **112** becomes the reference program side.

(91) When the reference program overtakes the setting side after the setting program sets the third pixel to the registers **801** and **802**, the register **803** refers to the information of the second pixel.

(92) In order to avoid such a situation, only the second pixel can be referenced by the reference program until the setting program finishes setting a third pixel in a configuration in a case where a double buffer method (registers **801** and **804**, registers **802** and **805**, and registers **803** and **806**) is taken, as in FIG. **8B**. After the setting program finishes writing the third pixel to the registers **801** to **803** and the reference program finishes referencing the second pixel from the registers **804** to **806**, the third pixel is transferred in one cycle.

(93) Although the configuration of FIG. **8B** is a common configuration, it requires twice as many flip-flops, resulting in a larger circuit scale.

(94) In the present embodiment, a shift register group is used as shown in FIG. **8C**. As a result, when all of the setter **104**, the first processing unit **110**, and the second processing unit **112** have completed one cycle of processing, the shift control unit **117** issues a shift cycle signal. The general-purpose shift register group **107** is used for calculation, and, depending on the application, it is possible to allocate a large number of interfaces between programs, or conversely, to allocate a large number of interfaces for circuit calculation. Therefore, it is possible to configure a programmable signal processing circuit having a large adaptive range with a small circuit.

(95) Although the first embodiment has been described above, the features thereof are summarized as follows.

(96) The programmable signal processor includes a general-purpose shift register group serially connected, a plurality of program execution units (a setter, a first processing unit, and a second processing unit are representative) that can execute in parallel with each other and use the general-purpose shift register group, and a shift control unit that generates a shift cycle signal for a shift request to the shift registers constituting the general-purpose shift register group and a start request for processing of one cycle in each program execution unit. In such a configuration, in a case where one computer program execution unit passes data to another program execution unit, the data is stored in the shift register one upstream of the shift register used by the other program execution unit for input. Also, in a case where one program execution unit passes a plurality of pieces of data to another program execution unit, data is stored in a plurality of shift registers positioned at intervals, sandwiching at least one shift register therebetween. In addition, in a case where an instruction indicating the end of processing for one cycle is detected, each program execution unit ends the operation and sets the program counter to the initial position in preparation for the next cycle. Then, the shift control unit issues a shift cycle signal in response to the detection of the end instruction of the processing for one cycle from all the program execution units.

(97) Note that the reading unit **101** does not use the general-purpose shift register group **107**. Also, the program executed by the reading unit **101** does not include an END instruction. This is because

the reading unit **101** stops reading in a case where the FIFO memory **103** is in the memory full state, and only needs to perform reading in a case where the memory is not in the memory full state.

(98) According to the above-described configuration, program execution units that can operate in parallel with each other can transfer data with each other by using a simple configuration called a serially connected shift register group, and can perform a given filter processing.

Second Embodiment

(99) A second embodiment will be described. The apparatus configuration is the same as that in FIG. **10** of the first embodiment, and description thereof will be omitted.

(100) FIG. **12** is a circuit configuration diagram of a programmable signal processor **1001** according to the second embodiment. The same components as in FIG. **1** are denoted by the same reference numerals.

(101) In the first embodiment, four program execution units—the reading unit **101**, the setter **104**, the first processing unit **110**, and the second processing unit **112**—operate in parallel. Generally, the greater the number of program execution units, the more the speed of processing of a signal processor can be improved, but the circuit scale also becomes accordingly more complex. In the second embodiment, there is no program execution unit corresponding to the second processing unit **112**, and the number of program execution units operating in parallel is reduced to further simplify the circuit scale. The processes of the reading unit **101**, the setter **104**, and the writing unit **114** are the same as those in the first embodiment.

(102) FIG. **13** shows a list of a program stored in the memory **111** of the first processing unit **110** according to the second embodiment. Since the first processing unit **110** of the second embodiment also includes and executes the program executed by the second processing unit **112** in the first embodiment, the number of program steps will increase. Since the throughput of the signal processor **1001** depends on the maximum number of program steps executed by the program execution unit, in the case of the configuration of the second embodiment, although the throughput is lower than that of the first embodiment, the circuit scale can be reduced.

Third Embodiment

(103) A third embodiment will be described. The apparatus configuration is the same as that in FIG. **10** of the first embodiment, and the description thereof will be omitted.

(104) FIG. **14** is a circuit configuration diagram of a programmable signal processor **1001** according to the third embodiment. The same components as in FIG. **1** are denoted by the same reference numerals.

(105) A feature of the signal processor **1001** in the third embodiment is that an interrupt signal generation register **1400** is added as shown in FIG. **14**. Although the interrupt signal generation register **1400** may be newly added, a register R1E which is one of the general-purpose registers **108** is used here. When “1” is written to bit **0** of the register R1E, the signal processor **1001** outputs an interrupt signal to an external unit (e.g., the CPU **1005**), causes the signal processor **1001** to transition to a suspended state, and stops until it is caused to resume by a control register from an external unit.

(106) The CPU **1005** performs predetermined interrupt processing by the interrupt signal. The predetermined interrupt processing performed by the CPU **1005** can be complex processing. Therefore, exceptionally complex processing by the signal processor **1001** can be taken over by the CPU **1005**, which dramatically increases the fields to which application is possible.

(107) In addition, when the CPU **1005** executes processing that does not include an arithmetic calculation, such as scanning a data string to detect a marker code, the total throughput, including the signal processor **1001**, can be improved since there is no overhead such as a loop jump instruction or a counter increment instruction.

Fourth Embodiment

(108) The fourth embodiment will be described. FIG. **15** is a block diagram of the apparatus in the

present fourth embodiment. In FIG. 15, a GPGPU unit (a general-purpose computing on GPU (Graphics Processing Unit)) **1501** is added to the configuration of FIG. 10.

(109) In an image capturing apparatus represented by a digital camera in recent years, autofocus is generally performed using a signal obtained from an image capturing surface. However, there is a problem in that, in order to convert a pupil-separated signal into a defocus amount, characteristics greatly change, depending on the coordinates on the image, which depend on optical characteristics of the sensor and optical characteristics of the lens. Conditions are even more complex in the case of a lens-interchangeable type image capturing apparatus.

(110) Such complex optical calculations use floating point polynomials. In view of this, in the fourth embodiment, the GPGPU unit **1501** obtains a defocus amount conversion coefficient. However, the side using the defocus amount conversion coefficient needs only about 8 bits of integer information, and it is necessary for the data to be processed to reduce the amount of data transferred.

(111) Therefore, in the fourth embodiment, the programmable signal processor **1001** is used for processing a result of a floating-point calculation.

(112) FIG. 17 shows a bit assignment of floating points. Bits are assigned to a sign part (positive or negative sign), an exponent part, and a mantissa part, respectively, and it is difficult to perform an auto-focus calculation at high speed as is.

(113) FIG. 18 is an equivalent circuit configuration diagram of the signal processor **1001** according to the fourth embodiment. A reference numeral **1800** corresponds to the setter **104**. Then, a register **1802** holds the floating point data to be set by the setter **104**. The shift operation transfers data from the register **1802** to the register **1803**. The CPU **1005** activates the signal processor **1001** after previously setting fixed values to the registers **R10** to **R16**. The values set in the registers **R10** to **R16** are as shown in the program list. Note, since there is no shift function from the register **R10** to the register **R1F**, the same value is held.

(114) FIG. 16 shows a list of a program stored in the memory **111** of the first processing unit **110** modeled after the circuit diagram of FIG. 18.

(115) In step **00**, the first processing unit **110** shifts the floating point of **R01**. Then, in step **01**, the first processing unit **110** performs mask processing to retrieve an exponent part. Then, in step **02**, the first processing unit **110** subtracts the offset for shifting a mantissa part.

(116) In step **03**, the first processing unit **110** masks the floating point data held in the register **R01** with a value preset in a register **R13**, and retrieves the mantissa part. Then, in step **04**, the first processing unit **110** adds the most significant bit of the mantissa part. This is because, in IEEE 754, the most significant bit of a mantissa part must always be 1.

(117) In step **05**, the first processing unit **110** shifts the mantissa part by the value held in the register **R02** to form an integer. Then, in step **06**, the first processing unit **110** subtracts the register **R03** from the value “0” set in advance in the register **R15**, thereby creating a negative value.

(118) In step **07**, the first processing unit **110** obtains the code of the floating point by logical multiplying the register **R01** holding the data of the floating point and the value held in the register **R16** in advance. Then, in step **09**, the first processing unit **110** selects a positive value or a negative value according to the sign, and writes the result to the output register **R1f**.

(119) While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

(120) This application claims the benefit of Japanese Patent Application No. 2022-099799, filed Jun. 21, 2022 which is hereby incorporated by reference herein in its entirety.

Claims

1. A signal processing apparatus comprising: a central processing unit (CPU); and a programmable signal processing circuit having the following: a plurality of execution circuits each of which has a program memory and executes a program stored in the program memory, wherein the plurality of execution circuits can operate in parallel; a register file used for the plurality of execution circuits, wherein the register file has a plurality of registers serially connected and transfers, in accordance with a shift signal, data held by each of the plurality of registers to a register located downstream; a shift controller that issues the shift signal to the plurality of execution circuits at a timing at which the plurality of execution circuits do not execute the program, wherein the CPU stores, to each program memory of the plurality of execution circuits, a program including instructions to make a register of the register file located upstream of a register that a second execution circuit of the plurality of execution circuits is to reference in order to input data be a storage destination for data of a result by execution of the program by a first execution circuit of the plurality of execution circuits so that the data of the result by the execution of the program by the first execution circuit is transferred to the second execution circuit via the register file.
2. The signal processing apparatus according to claim 1, wherein the shift controller issues the shift signal to the plurality of execution circuits in a case where each of the plurality of execution circuits has completed execution of one cycle of the program.
3. The signal processing apparatus according to claim 2, wherein each of the plurality of execution circuits stops executing the program in a case where each of the plurality of execution circuits has completed execution of one cycle of the program and executes the program again in accordance with the shift signal.
4. The signal processing apparatus according to claim 2, wherein each of the execution circuits has a program counter and a detector that detects an instruction indicating an end of one cycle of the program, wherein the program counter stops updating a count value in accordance with a detection signal from the detector and sets a head address of the program to the program counter, and wherein the shift controller issues the shift signal in a case where the shift controller has received the detection signals from the detectors of all of the plurality of execution circuits.
5. The signal processing apparatus according to claim 1, wherein the execution circuit includes a setting circuit that sets data to be processed to the register, wherein the programmable signal processing circuit has a reading circuit that has a program memory, wherein the reading circuit, by executing a program stored in the program memory of the reading circuit, reads the data to be processed from an external memory and outputs the read data to the setting circuit, and the CPU stores a program for specifying the data to be processed to the program memory of the reading circuit.
6. The signal processing apparatus according to claim 5, wherein the reading circuit outputs the data to be processed to the setting circuit via a FIFO memory, and wherein the reading circuit, in a case where the FIFO memory has become full, stops reading the data to be processed from the external memory, and in a case where free space has been generated in the FIFO memory, resumes reading.
7. The signal processing apparatus according to claim 1, wherein the register files has an output register that stores data of a result of a predetermined process by the programmable signal processing circuit, and wherein the CPU stores, to the program memory of an execution circuit that outputs the data of a result of a predetermined process among the plurality of execution circuit, a program including an instruction for storing data obtained by executing the program.
8. The signal processing apparatus according to claim 1, wherein the program memory of each of the plurality of execution circuits and each register of the register file are mapped to different address spaces in a memory space of a predetermined IO bus.
9. The signal processing apparatus according to claim 1, wherein the register file has a plurality of general-purpose registers that hold data unrelated to the shift signal, wherein the programmable

- signal processing circuit shifts to a suspended state and an interrupt signal is output to an outside in a case where a predetermined value is stored in a predetermined one of the plurality of general-purpose registers.
10. The signal processing apparatus according to claim 1, wherein the register file has a plurality of general-purpose registers that hold data unrelated to the shift signal, wherein the CPU sets a predetermined value on the general-purpose register and stores a program including an instruction for using the general-purpose register to the program memory of the plurality of execution circuits.
11. The signal processing apparatus according to claim 1, wherein the CPU reads the program executed by each of the plurality of execution circuits from a memory, and stores the read program to the program memory of each of the plurality of execution circuits.
12. The signal processing apparatus according to claim 11, wherein the CPU executes the program stored in the memory and controls the image processing apparatus.
13. A non-transitory computer-readable storage medium storing programs to be executed by a signal processing apparatus comprising: a central processing unit (CPU); and a programmable signal processing circuit having the following: a plurality of execution circuits each of which has a program memory and executes a program stored in the program memory, wherein the plurality of execution circuits can operate in parallel; a register file used for the plurality of execution circuits, wherein the register file has a plurality of registers serially connected and transfers, in accordance with a shift signal, data held by each of the plurality of registers to a register located downstream; a shift controller that issues the shift signal to the plurality of execution circuits at a timing at which the plurality of execution circuits do not execute the program, wherein the program stored in the storage medium causes the CPU to store, to each program memory of the plurality of execution circuits, a program including instructions to make a register of the register file located upstream of a register that a second execution circuit of the plurality of execution circuits is to reference in order to input data be a storage destination for data of a result by execution of the program by a first execution circuit of the plurality of execution circuits so that the data of the result by the execution of the program by the first execution circuit is transferred to the second execution circuit via the register file.
14. The signal processing apparatus according to claim 1, wherein the signal processing apparatus is an image capturing apparatus, and wherein the programmable signal processing circuit processes an image signal.
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